

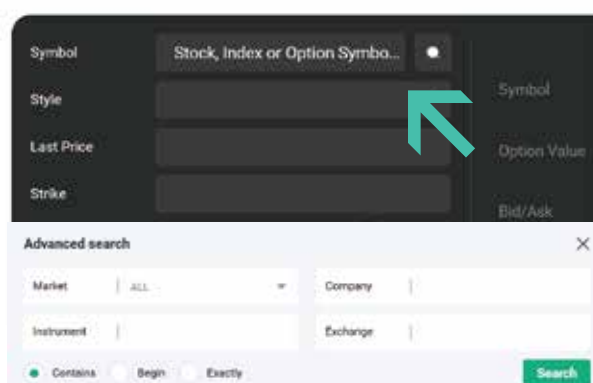
Options Calculator

Powered by IVolatility

The Options Calculator, powered by IVolatility.com, provides fair values and Greeks of any option using IVolatility.com data and 20-minute delayed prices. You may customize all the input parameters (option style, price of the underlying instrument, strike, expiration, implied volatility, interest rate and dividends data) or enter a stock or options symbol and the database will populate all fields for you. Note that the option's underlying price is 20-minute delayed. If the markets are closed, the previous trading day's market closing price will be used.

Choose a Stock Symbol, Index Symbol, or Option Symbol

Input any stock, index or option symbol. If you don't know the symbol or the option you can browse by company name, instrument, market or exchange via the "Symbol Lookup" icon. Once you select a symbol, data will populate the table with the stock's price and the closest to expiring at-the-money options.



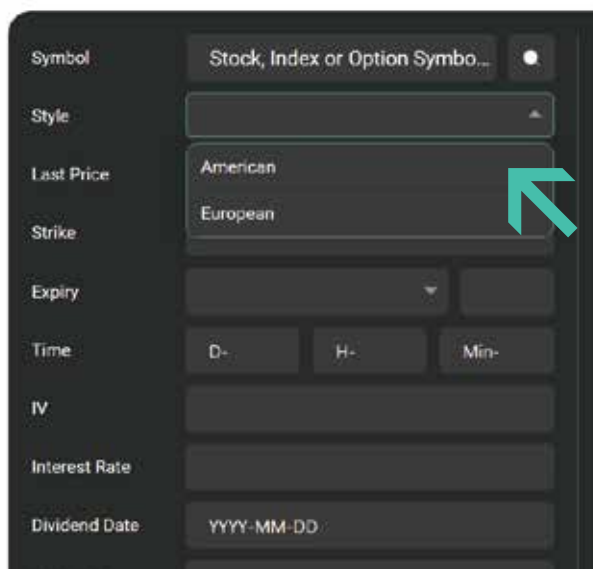
The screenshot shows the top part of the Options Calculator interface. It features a 'Symbol' input field with a placeholder 'Stock, Index or Option Symbo...' and a magnifying glass icon. To the right of this field is a 'Symbol Lookup' icon, represented by a magnifying glass with a red arrow pointing to it. Below the main input fields is an 'Advanced search' section with filters for Market, Company, Instrument, and Exchange, along with radio buttons for 'Contains', 'Begin', and 'Exactly', and a 'Search' button.

American-style vs. European-style

Next, select whether to price the option as an American-style or European-style type option.

AMERICAN-STYLE: The majority of stock options listed for trading on the options exchanges in the United States are American-style options. This means that the option can be exercised at any time on or before expiration. This early exercise capability is factored into the pricing of the options. There are some index options that are American-style and you should check the product specifications supplied by the exchange for this information. This Options Calculator uses a binomial model with 100 steps to create a theoretical price for American-style options.

EUROPEAN-STYLE: Most (but not all) index options listed for trading on the options exchanges in the United States are European-style options. This means that the option can be exercised only on its expiration date. This inability to exercise until expiration is factored into the pricing of the options. This option calculator uses a Black-Scholes model for European-style options.



The screenshot shows the 'Style' input field in the Options Calculator interface. It is a dropdown menu with 'American' and 'European' options. A red arrow points to the dropdown arrow on the right side of the field. Below the style field are other input fields for 'Last Price', 'Strike', 'Expiry', 'Time', 'IV', 'Interest Rate', and 'Dividend Date'.

Last Price

The option's underlying price is 20-minute delayed. If the markets are closed, the previous trading day's market closing price will be used. The user can change it to today's current price or any other price by manually entering the desired price. Notice that the price must be entered in decimal format.

Strike

The strike price of the option shown can be changed manually. The default is the at-the-money strike.

Expiry

This will populate a drop down with all potential expiration dates. The default value will be the closest expiry date, or the option with the fewest days to expiration. Expiry date and time are customizable and modifications will display possible impact to an option's value.

Volatility

The default Volatility is automatically populated and derived from last night's implied volatility for that particular put and call option series. Users can change the volatility by entering any other value.

Interest Rate

The default risk-free interest rate is derived from last night's treasury market and is the rate that is equivalent to the option's expiration term. Users can change the interest rate by entering any other value.

Dividend Amount, Date & Frequency

Depending upon whether a stock pays a dividend or not, this section may or may not contain data. If there is data within the fields you can alter it by simply adding in the dividend amount and the corresponding ex-dividend date into the fields. The calculator will then price the options accordingly.

Symbol: Stock, Index or Option Symbol...
Style: [dropdown]
Last Price: [input box]
Strike: [input box]
Expiry: [dropdown]

Symbol: [input box]
Style: American
Last Price: 217.41
Strike: 217.5

Expiry: 2023-11-03
Time: 16:00
Time dropdown: 2023-11-03, 2023-11-10, 2023-11-17, 2023-11-24, 2023-12-01, 2023-12-08
IV: 6
Interest Rate: [input box]
Dividend Date: [input box]

Time: 1, 4, 6
IV: 52.87
Interest Rate: 5.3500

IV: 52.87
Interest Rate: 5.3500
Dividend Date: [input box]

Dividend Date: YYYY-MM-DD
Dividend Amount: 0
DivFreq: [dropdown]

Calculation

At this step, the user is now ready to calculate the call and put prices for this option. Using the values entered, the Greeks are calculated in conjunction with the options prices. The Greeks (Delta, Gamma, Theta, Vega and Rho) are mathematical values that measure the sensitivity of an option's price to stock, time, volatility and interest rate changes.

DELTA: The theoretical amount an option's price will change for a corresponding one-point change in the price of the underlying security.

GAMMA: The theoretical change in Delta divided by the dollar change in the price of the underlying security's price. It is the measurement of the rate of change in an option's Delta with respect to the underlying price.

THETA: A measure of how much an option's price theoretically decays with each passing day with the price of the underlying security and implied volatility remaining unchanged.

VEGA: The theoretical measure of change in an option's price in response to a one percentage point change in implied volatility.

RHO: A measure of change in an options price in response to a one percentage change in the risk-free interest rate.

ALPHA: Alpha is a ratio of Gamma over Theta. Thus Alpha indicates the relative value of owning gamma relative to the current level of theta. It is a measure that allows for comparison of several different options on the basis of how much they cost daily to own (daily Theta) versus the potential gamma derived return (profits from movement) from owning them. The greater absolute value of Alpha the more potential for profits exists against the loss from Theta for long positions. The converse is true for short positions.

Options Price & Implied Volatility

On the bottom, left side of the tool, the user has the ability to input a different options price. When this is done, the user will notice the corresponding implied volatility reading that would be tied to that option price with all other factors remaining unchanged. In this, users may be able to perform a "what if" simulation to view how option prices might be affected by larger changes in implied volatility levels, with all other factors remaining static.

	Call	Put
Symbol	TSLA 231103C0021...	TSLA 231103P0021...
Option Value	2.1340	2.2005
Bid/Ask	2.53/ 2.56	2.61/ 2.64
Delta	0.5003	-0.5001
Gamma	0.0735	0.0739
Theta	-1.3461	-1.3210
Alpha	-0.0546	-0.0559
Vega	0.0410	0.0410
Rho	0.0024	-0.0022

Symbol	TSLA 231103C0021...	Symbol	TSLA 231103C0021...	Put	TSLA 231103P0021...
Style	American	Option Value	2.3944		1.6483
Last Price	216.23	Bid/Ask	2.72/ 2.76		2.87/ 2.91
Strike	217.5	Delta	0.5643		-0.4360
Expiry	2023-11-09	Gamma	0.0796		-0.0789
Time	1	Theta	-1.4308		-1.4252
IV	50.87	Vega	-0.0542		-0.0504
Interest Rate	3.50%	Alpha	0.0373		0.0373
Dividend Date	YYYY-MM-DD	Rho	0.0023		-0.0017
Dividend Amount	0	Implied Volatility	50.0		60.0
Call/Put	Call				
Option Price	218				

Disclaimer

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